

HPC User support and Onboarding

Solutions for HPC: Best Practices and Common Sense

Tuesday, August 4, 2026

SDSC
SAN DIEGO SUPERCOMPUTER CENTER

UC San Diego

Agenda

- HPC Ecosystem
- Available Help Resources
- Common Issues & solutions
 - Login, file systems, performance, batch scheduler, allocations
- Getting an allocation of your own
 - New NSF requirements
- Take Aways
- Resources

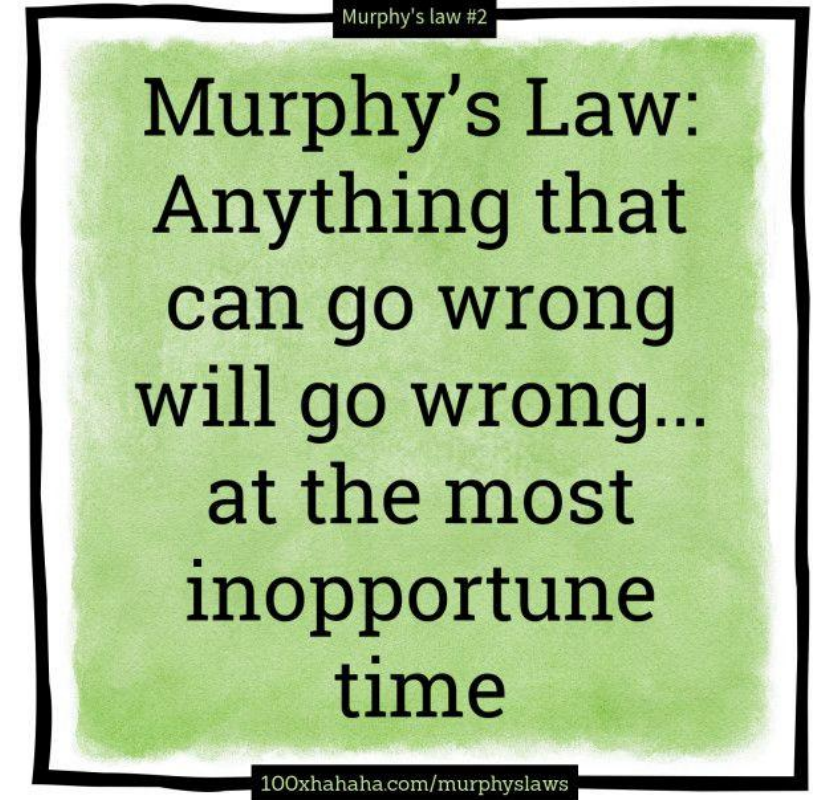
HPC Ecosystem

- You are not alone (HPC is a shared resource)
 - As a user (HPC ecosystem)
 - In getting support (available resources)
- What all is shared
 - Login Nodes
 - File systems
 - Compute nodes
 - System Administration, Security, Networking and System support teams
 - System policies



General Tips

- Plan Ahead
 - Don't leave HPC tasks to the last minute
- Design for recovery
 - Back up your data
 - Use version control



Best Practices: Getting help

- Understand your problem
- Engage with appropriate support tools
- Provide relevant and adequate information for helpdesk to reduce iterations
- Always be nice to the support desk! 😊

What are the available Help options?

- What are the available resources(you are not alone)
 - User
 - Project/Colleagues
 - Web (Community forums, User guides, Git repositories)
 - Helpdesk

Types of HPC Support Available

- Training workshops and tutorials
- User forums and community channels
- Documentation and knowledge bases
- Help desk or ticketing system
- Office hours and consultation services
- Research computing consultants

Training and Educational Resources

- User onboarding materials
- Workshops and webinars
- Example job scripts
- Best practices guides
- Self-paced learning resources
 - <https://www.sdsc.edu/education/on-demand-learning/index.html>

Before Asking for Help

- Check system status pages
- Review documentation
- Search existing FAQs and knowledge bases
- Examine job logs and error messages
- Verify account allocations and quotas
- Test with a small job

How to Submit an Effective Support Request

- Include:
 - User ID/account name
 - Cluster/system name
 - Job ID(s)
 - Error messages
 - Commands or scripts used
 - Software versions/modules
 - Steps to reproduce the problem
 - Time the issue occurred
 - Example: "Job 123456 on Cluster X failed with a memory error after running for 2 hours. Attached are the job script and output logs."

HPC Support Etiquette and Expectations

- Recall you are not alone.....
- Response times and service levels
- Prioritization of issues
- Following up on tickets
- Sharing resolutions with collaborators

Common Issues

- Login and Account problems
- Jobs/Policies/Schedulers
- Allocations/Job charging
- Applications/software
- Storage, Data storage, and Data Sharing
- Performance

Best Practices: Secure your credentials

- Passwords
 - Don't reuse passwords
 - Longer is better
 - Don't keep digital plaintext copies of passwords
 - Don't Hard code password in files
 - Don't share passwords
- Use password-manager program
- Use SSH keys, ssh-agent
- Multi-factor Authentication(MFA)

Common Issues: Login and account issues

- Access (first time)
 - Different resources have different access protocols, or mechanisms for access, even if its at the same site
 - Direct login
 - Portals
 - Password and username issues
 - Problem: Indicator message: Enter verification code
 - Solution: check username <https://allocations.access-ci.org/profile> (if username is not available for the resource then the account has not been created yet) Generally it can take 1 business day for accounts to be fully functional.
 - ssh keys
 - Problem: Indicator message: Enter password
 - Solution: resend ssh pub key, or use -i option to point to location of private key
 - MFA(2FA)

Common Issues: Resource Access/Logging In

- Unable to access system(returning user)
 - Cant log in
 - Problem: System Maintenance
 - Solutions:
 - Check specific error message
 - User News
 - Stay Subscribed to be notified
 - Check on User Portal -- <https://support.access-ci.org/announcements>
 - Hanging on login
 - Problem: Users overstimulating the file system
 - Solution: Support staff contact users to change behaviors
 - Problem: Activating Conda in your .bashrc
 - Solution: Do not initialize conda in .bashrc, but only when needed in run scripts

Common Issues: Job Submissions

- Submitting Jobs

- Missing Allocation
- Missing Software/License
- Improper QOS
- Job Scripts
 - Allocations
 - Queues/Partitions
 - Time Limits
 - OOM
 - New Software stack
 - Module spider
 - Expanse-client tool
 - Error message
 - sbatch: error: Project balance is not enough to run the job
 - sbatch: error: QOSMaxNodePerJobLimit
 - sbatch: error: Batch job submission failed: Job violates accounting/QOS policy (job submit limit, user's size and/or time limits)

```
login01 user_support]$ sbatch azton.sb  
sbatch: error: Batch job submission failed: Invalid account or  
account/partition combination specified
```

```
login01 user_support]$ sbatch azton.sb  
sbatch: error: QOSMaxCpuPerJobLimit  
sbatch: error: Batch job submission failed: Job violates accounting/QOS  
policy (job submit limit, user's size and/or time limits)
```

Best Practices: jobs and job charging

- Check user guide for accounting policies
- Use system tools for most up to date accounting information
 - Scheduler tools: Slurm for individual job details
 - squeue
 - sacctmgr
 - scontrol
 - sinfo
 - sacct
 - Home grown tools for accounting information
 - Expanse-client

squeue

- Reports job status and reason codes
 - Queued Jobs (*ReqNodeNotAvail, Reserved for maintenance)

```
[nickel@login01 ~]$ squeue | more
```

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	NODELIST (REASON)
13574113	compute	80dgree_	yweng3	PD	0:00	2	(MaxMemPerLimit)
12668967	compute	0-xtensi	kavousi	PD	0:00	1	(MaxMemPerLimit)
14756880	compute	job001_p	amysai	PD	0:00	10	(Reservation)
14800161	compute	namd-com	sasadian	PD	0:00	6	(QOSMaxCpuPerUserLimit)
14800218	compute	namd-com	sasadian	PD	0:00	6	(QOSMaxCpuPerUserLimit)
14789098	compute	bl_8JHNp	uscms	PD	0:00	1	(MaxJobsPerAccount)

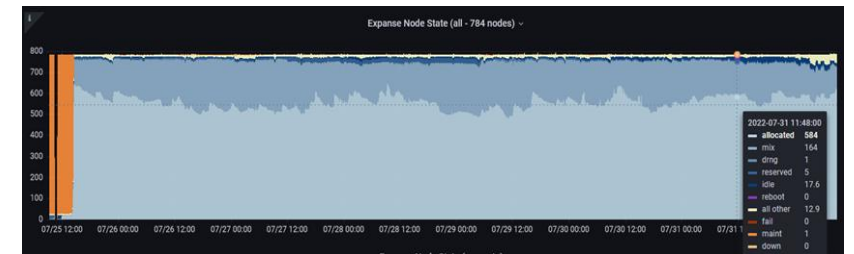
- Running jobs

14813206	compute	post0110	lpegolot	R	16:30:28	1	exp-9-35
14800090	compute	namd-com	sasadian	R	16:13:01	6	exp-2-29,exp-3-23,exp-4-33,exp-7-20,exp-9-[03,26]
14764467	compute	V1WTReRU	aminkvh	R	16:08:56	1	exp-2-54
14773832	compute	V4R1639Q	aminkvh	R	15:55:58	1	exp-8-14
14800092	compute	namd-com	sasadian	R	15:29:28	6	exp-4-29,exp-7-[07,39-40],exp-9-[28,41]
14812166	compute	scratch	mlaskow2	R	15:53:59	1	exp-10-20
14812167	compute	scratch	mlaskow2	R	15:39:34	1	exp-8-48
14800158	compute	namd-com	sasadian	R	15:17:18	6	exp-2-[26,50],exp-4-[52-53],exp-7-[42-43]
14812168	compute	scratch	mlaskow2	R	15:20:01	1	exp-10-37

Common Issues: Job Submissions

- **squeue – Common “reasons” for pending jobs**
 - MaxMemPerLimit
 - Max. mem per Node = 243G
 - QOSMaxNodePerUserLimit
 - Priority
 - ReqNodeNotAvail, Unavailable nodes: exp-x-xx
- **File system not available**
 - We have added a slurm directive `#SBATCH –constraint = “lustre”` to indicate if your job is using the lustre file system. If this is provided, the scheduler will not launch the job on a node that is missing lustre.
- **System Maintenance**
 - <https://support.access-ci.org/outages>

sacctmgr



- View or modify Slurm account information
- `sacctmgr show qos`
 - Lists policies on queue/partitions, wait times, Time Limits
- User Guide

```
login01 ~]$ sacctmgr show qos format=name%20,MaxWall,MaxTRESPU%20,MaxJobsPU,MaxSubmitPU,MaxTRESPA%20,MaxJobsPA,MaxSubmitPA
```

Name	MaxWall	MaxTRESPU	MaxJobsPU	MaxSubmitPU	MaxTRESPA	MaxJobsPA	MaxSubmitPA
normal	2-00:00:00	cpu=8192,node=64	32	64	cpu=16384,node=128	32	64
shared-normal	2-00:00:00	cpu=8192,node=64	4096	4096	cpu=16384,node=128	4096	4096
large-shared-normal	2-00:00:00		2	4		4	4
preempt-normal	7-00:00:00			128	cpu=4096,node=32		128
gpu-normal	2-00:00:00	cpu=160,gres/gpu=16+	4	8	gres/gpu=32,node=8	8	16
gpu-shared-normal	2-00:00:00	cpu=240,gres/gpu=24+	24	24	cpu=320,gres/gpu=32+	24	24
gpu-preempt-normal	7-00:00:00	gres/gpu=24,node=6	12	16	gres/gpu=48,node=12	16	20
debug-normal	01:00:00	cpu=256,node=2	2	2	cpu=256,node=2	2	2
gpu-debug-normal	00:30:00	gres/gpu=8,node=2	2	2	gres/gpu=8,node=2	2	2

Common Issues: Queue and Time Limits

Partition Name	Max Walltime	Max Nodes/Job	Max Running Jobs	Max Running + Queued Jobs	Charge Factor	Notes
compute	48 hrs	32	32	64	1	Exclusive access to regular compute nodes; <i>limit applies per group</i>
ind-compute	48 hrs	32	32	64	1	Exclusive access to Industry compute nodes; <i>limit applies per group</i>
shared	48 hrs	1	4096	4096	1	Single-node jobs using fewer than 128 cores
ind-shared	48 hrs	1	32	64	1	Single-node Industry jobs using fewer than 128 cores
gpu	48 hrs	4	4	8 (32 Tres GPU)	1	Used for exclusive access to the GPU nodes
ind-gpu	48 hrs	4	4	8 (32 Tres GPU)	1	Exclusive access to the Industry GPU nodes
gpu-shared	48 hrs	1	24	24 (24 Tres GPU)	1	Single-node job using fewer than 4 GPUs
ind-gpu-shared	48 hrs	1	24	24 (24 Tres GPU)	1	Single-node job using fewer than 4 Industry GPUs
large-shared	48 hrs	1	1	4	1	Single-node jobs using large memory up to 2 TB (minimum memory required 256G)
debug	30 min	2	1	2	1	Priority access to shared nodes set aside for testing of jobs with short walltime and limited resources
gpu-debug	30 min	2	1	2	1	Priority access to gpu-shared nodes set aside for testing of jobs with short walltime and limited resources; <i>max two gpus per job</i>
preempt	7 days	32		128	.8	Non-refundable discounted jobs to run on free nodes that can be pre-empted by jobs submitted to any other queue
gpu-preempt	7 days	1		24 (24 Tres GPU)	.8	Non-refundable discounted jobs to run on unallocated nodes that can be pre-empted by higher priority queues

https://www.sdsc.edu/support/user_guides/expanse.html#running

scontrol

- View or modify Slurm configurations and state on currently queued or running jobs
- scontrol [OPTIONS...] [COMMAND...]
- Show individual job information
 - scontrol show job <<local_jobid>>

```
login01]$ scontrol show job 32658556
JobId=32658556 JobName=NGBW-JOB-BEAST2_XSEDE-43842C4A2D8C4B5085E71285ADC4D5DB
  UserId=testuser(505687) GroupId=use300(6099) MCS_label=N/A
  Priority=4509 Nice=0 Account=sds121 QOS=shared
  JobState=RUNNING Reason=None Dependency=(null)
  Requeue=1 Restarts=0 BatchFlag=1 Reboot=0 ExitCode=0:0
  RunTime=00:46:22 TimeLimit=03:00:00 TimeMin=N/A
  SubmitTime=2024-07-30T12:22:15 EligibleTime=2024-07-30T12:22:15
  AccrueTime=2024-07-30T12:22:15
  StartTime=2024-07-30T12:22:16 EndTime=2024-07-30T15:22:16 Deadline=N/A
  SuspendTime=None SecsPreSuspend=0 LastSchedEval=2024-07-30T12:22:16
Scheduler=Main
  Partition=shared AllocNode:Sid=login01:2295441
  ReqNodeList=(null) ExcNodeList=(null)
  NodeList=exp-16-53
  BatchHost=exp-16-53
  NumNodes=1 NumCPUs=1 NumTasks=1 CPUs/Task=1 ReqB:S:C:T=0:0:*:*
  ReqTRES=cpu=1,mem=1G,node=1,billing=3600
  AllocTRES=cpu=1,mem=1G,node=1,billing=3600
  Socks/Node=* NtasksPerN:B:S:C=1:0:*:* CoreSpec=*
  MinCPUsNode=1 MinMemoryNode=1G MinTmpDiskNode=0
  Features=(null) DelayBoot=00:00:00
  Reservation=cipres-shortjobs
  OverSubscribe=OK Contiguous=0 Licenses=cipres:1 Network=(null)
  Command=._batch_command.run
  WorkDir=/exppanse/projects//NGBW-JOB-BEAST2_XSEDE-
43842C4A2D8C4B5085E71285ADC4D5DB
  StdErr=/exppanse/projects//NGBW-JOB-BEAST2_XSEDE-
43842C4A2D8C4B5085E71285ADC4D5DB/_scheduler_stderr.txt
  StdIn=/dev/null
  StdOut=/exppanse/projects//NGBW-JOB-BEAST2_XSEDE-
43842C4A2D8C4B5085E71285ADC4D5DB/_scheduler_stdout.txt
  Power=
  MailUser=user@sdsc.edu
MailType=INVALID_DEPEND,BEGIN,END,FAIL,REQUEUE,STAGE_OUT
```

Sinfo: Why is my job not running? Queue, wait times, Time Limits

- View information about nodes and partitions

```
login01 ~]$ sinfo
PARTITION      AVAIL  TIMELIMIT  NODES  STATE NODELIST
compute        up 2-00:00:00      1  inval exp-1-15
compute        up 2-00:00:00      2 drain$ exp-13-[55-56]
compute        up 2-00:00:00      3 drain* exp-2-07,exp-5-50,exp-9-23
compute        up 2-00:00:00      7  drng  exp-3-30,exp-4-17,exp-5-
[18,29,39,53],exp-7-45
compute        up 2-00:00:00      1  drain exp-4-55
compute        up 2-00:00:00     12  resv  exp-2-[21-24],exp-14-50,exp-16-
[54-56],exp-17-[53-56]
compute        up 2-00:00:00    230   mix  exp-1-[01-03,12-14,16,18,20,22-
26,28-39,41,43-44,47-48,51-54,56],
```

https://www.sdsc.edu/support/user_guides/expanse.html#running

sacct

- View accounting data for completed jobs and job steps
- Usage: `sacct [OPTIONS...]`

```
[nickel@login01 ~]$ sacct -j 28582088 -l
JobID      JobIDRaw      JobName      Partition  MaxVMSize  MaxVMSizeNode  MaxVMSizeTask  AveVMSize      MaxRSS  MaxRSSNode  MaxRSSTask
AveRSS  MaxPages  MaxPagesNode  MaxPagesTask  AvePages      MinCPU  MinCPUNode  MinCPUTask      AveCPU      NTasks  AllocCPUS      Elapsed
State  ExitCode  AveCPUFreq  ReqCPUFreqMin  ReqCPUFreqMax  ReqCPUFreqGov      ReqMem  ConsumedEnergy  MaxDiskRead  MaxDiskReadNode  MaxDiskReadTask
AveDiskRead  MaxDiskWrite  MaxDiskWriteNode  MaxDiskWriteTask  AveDiskWrite      ReqTRES  AllocTRES  TRESUsageInAve  TRESUsageInMax
TRESUsageInMaxNode  TRESUsageInMaxTask  TRESUsageInMin  TRESUsageInMinNode  TRESUsageInMinTask  TRESUsageInTot  TRESUsageOutMax  TRESUsageOutMaxNode
TRESUsageOutMaxTask  TRESUsageOutAve  TRESUsageOutTot
```

- `sacct -j 123456 format=jobid,partition%20`
- For completed jobs it is encouraged to include where the job files are located including submit script, `.err`, and `.out` files

Common Issues: Accounting/Charging

- All systems charge differently
- ACCESS allocates in ACCESS Credits which can be converted to SUs (service Unit)
 - Each resource has a unique definition of an SU
 - Some are in core hours, some are in node hours, etc
 - Some charge for other components such as memory
- Visit the user guide for specifics
 - Check for opportunities to save
- Allocations are shared
- Charging is generally based on what is requested, not how resources are used
- Do test jobs to evaluate
 - Slurm commands to collect information
 - `sacct -u $USER`
 - `sacct -j $JOBID`

For Expanse Specifically.....

- Charging is based on what is requested, not how resources are used
- Charging is based on the Maximum of memory and CPU core fraction
- Minimum charge for any job is 1SU

Example for CPU

$\text{Max} [3600 * \# \text{CPU cores}, 1800 * \# \text{Mem in GB}] / 3600 * (\text{wallclock time in secs} / 3600)$

- 1 CPU and less than 2GB of memory are charged 1 CPU Service Unit (1SU = 1 core/hour).
- 1 GPU and up to 10 CPUs and 92 GB of memory are charged 1 GPU Service Unit (SU)/hour. This job will be charged 1 GPU SU/hour.
- The minimum charge for any job is 1 SU.
- 1 Expanse SUs = 1 ACCESS Credit
- 1 Expanse GPU SU = 54 Expanse SUs (for conversion)
 - https://allocations.access-ci.org/exchange_calculator

Accounting

- SDSC uses slurm as a scheduling tool for cluster management and accounting
- Different resources use different home grown tools to help users evaluate their usage.

- Expanse-client tool(SDSC)
- Projects (PSC)
- taccinfo (TACC)

```
login01 ~]$ expanse-client user train112 -p
```

```
Resource expanse
```

NAME	STATE	PROJECT	TG PROJECT	USED	AVAILABLE	USED BY PROJECT
------	-------	---------	------------	------	-----------	-----------------

train112	allow	gue998	TG-CIE960001S	337	200000	114621
----------	-------	--------	---------------	-----	--------	--------

- ACCESS Portal updated at various intervals

Common Issues: Software

- Software Installs
 - Modules
 - Compile (Build your own)
 - Containers
- Most HPC system use module to manage their software stacks
- Login nodes, compute nodes are different
 - Compiling codes need to happen on the nodes on which they will run
 - Modules need to be loaded on the nodes on which applications will run
- Software managed by the resource provider has usually been tested and fine tuned for the specific resources

Best Practices: Software

- Review user guide for tools available
- Use system installed applications when available
- Use containers to manage out of date software

Environment Modules

- Environment Modules help manage software incompatibilities, versioning, and dependencies
- Environment Modules provide for dynamic modification of your shell environment
- Module commands set, change, and/or delete environment variables
- Modules manage software versions
- Module manage dependencies by loading or unloading other modules
 - Check for dependencies with module spider <module_name>
- Module list - lists all the currently loaded modules

Navigating with Modules

Command	Description
<code>module list</code>	Currently Loaded Modules
<code>module avail</code>	List of available software modules based on your current module path
<code>module spider</code>	List all available software and versions on the system
<code>Module spider <application name></code>	List available application specific modules and module details, including versions and dependencies
<code>module load [module file]</code>	Load module(s) or specify unresolved dependencies
<code>module show [module file]</code>	Display information about loaded modules including changes, dependencies, versions and paths
<code>module unload [module file]</code>	Unloads a specified module form the environment
<code>module purge</code>	Unloads all the loaded modules
<code>module reset</code>	Reset modules to default settings

Common Issues: File systems

- Cant write to home
- Sharing
- File system hanging
- Backups

What is in a file system

- Common File systems and their utilization
 - Home – Usually limited in space
 - Scratch - Large space, limited persistence, no backup
 - Node local scratch(SSD)- good performance, only available during job
 - Persistent storage – good performance, limited persistence
 - Archival- Slow, backup
- Types of file systems
 - Lustre
 - NFS
 - GPFS
 - Ceph

File management

- Review file permissions `ls -la`
- Review usage
 - `ls`
 - `du`
- Review user group
 - `Groups username, id username`
- Controls: Permissions granularity levels
 - (Attribute), User, Group, Other
 - Read(4), Write(2), Execute(1)
 - Default 755 (User(read, write, execute):Group(read, execute): Other(read, execute))
 - Use `chmod`, `chown` commands to modify ownership and permissions
 - X is capitalized so that group execute permission is given to any files that are executables (don't use lower case because that would change all files to have execute permissions).

Data Management & Backup Best Practices

- Users are responsible for their HPC data.
- Plan transfers; archive small files before moving them.
 - Transfers take time
- Back up critical data regularly; keep a few known-good versions.
- Back up only essential, hard-to-recreate files (code, scripts, configs, key outputs).
- Use Git for version control.
- Verify transfers with checksums.
- Store backups on a separate, secure resource and test restores.
- Follow HPC file system and retention guidelines.

Common Issue: system performance

- GPU
- CPU
- Login Nodes
- Lustre

Monitoring Resource Utilization

- Tools
 - top, htop, atop : display Linux processes
 - mpstat : display processors related statistic
 - sar : display system activity report
 - free: display memory statics
 - ps: display active processes
 - nvidia-smi: display gpu activity and statistic
- seff (slurm efficiency) after job is complete

Allocations

- SDSC allocates resources via two methods: peer review process or for purchase via three different programs: NSF allocated resources (ACCESS-CI/NAIRR), Discretionary time (HPC@UC), and TSCC & Industrial program.
 - ACCESS-CI: (Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support) (<https://access-ci.org/>)
 - NAIRR: (National AI research resource) (<https://nairrpilot.org/>)
 - HPC@UC: Available to UC folks to gain quick access to limited compute and storage resources with goal to graduate to ACCESS proposal
 - TSCC (Triton Shared Compute Cluster): Purchase resources on our Condo/Hotel cluster
 - Industrial Partners: Purchase SUs on our Expanse Industry Rack

ACCESS: Types of ACCESS Allocations

- **Trial allocation: 100 GPU and/or 1000 CPU hours**
 - **Submit request to:** consult@sdsc.edu
- **Explore Allocations**
 - Small, fast-turnaround requests
 - Ideal for testing and new users
- **Discover Allocations**
 - Startup projects and educational activities
- **Accelerate Allocations**
 - Medium-sized research projects
- **Maximize Allocations**
 - Large-scale computational research projects
- **Apply for allocations through ACCESS**
 - <https://allocations.access-ci.org/prepare-requests-overview>

Allocation	Credit Threshold
<u>Explore ACCESS</u>	400,000
<u>Discover ACCESS</u>	1,500,000
<u>Accelerate ACCESS</u>	3,000,000
<u>Maximize ACCESS</u>	Not awarded in credits.

HPC@UC

- Request HPC@UC at: https://www.sdsc.edu/support/hpc_uc_apply-exp.html
- Up to 500K core-hours of computing, associated data storage, and access to SDSC expertise to assist their research team.
- Awards are active for one year. NO supplements, renewals or Extensions
- Applicants must not have an active ACCESS award
- Developed to support onboarding to ACCESS and larger, formal allocation requests
- SDSC staff will assist in developing these allocation applications.
- Applications are reviewed on an ongoing basis. Applicants will be notified within 10 business days of the review decision.
- https://www.sdsc.edu/collaborate/hpc_at_uc.html

Writing a Strong Allocation Request

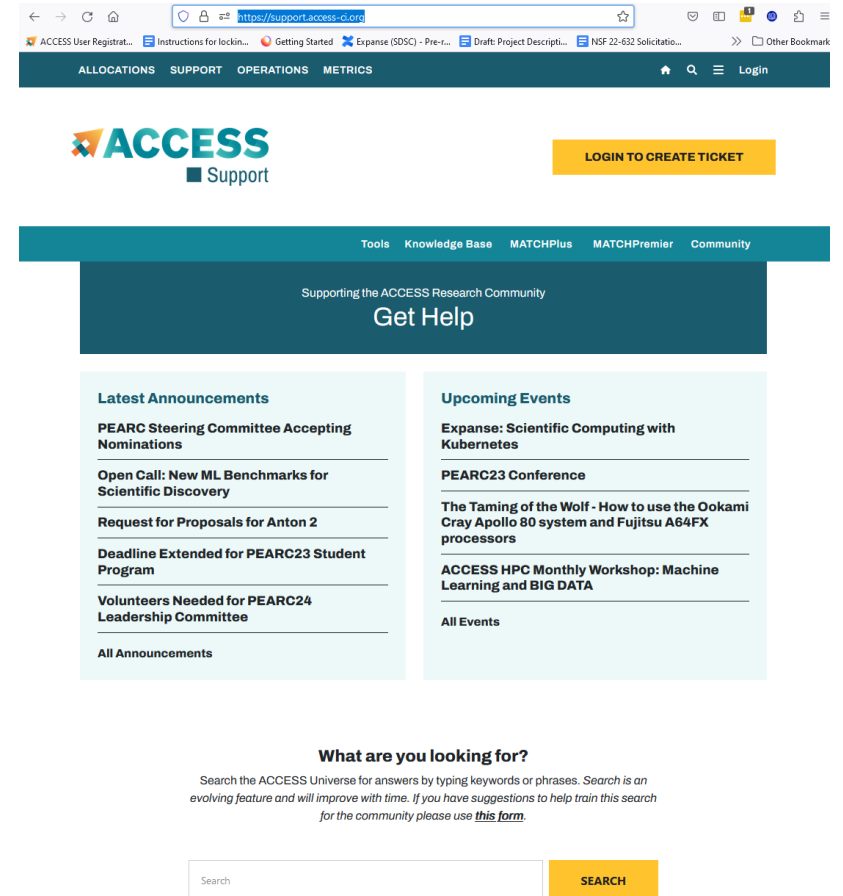
- **Resource Justification**
- Explain:
 - Number of CPU/GPU hours needed
 - Storage requirements
 - Software requirements
 - Scaling tests (if available)
- **Common Mistakes**
 - Overestimating resources
 - Vague project descriptions
 - Missing computational methodology
 - Insufficient justification

Key Takeaways

- You are not alone
- Lots of resources available
- Lots of common issues that are well documented
- Take advantage of available documentation and training
- When using support
 - Collect relevant information before contacting support
 - Provide complete details in support requests

Resources

- ACCESS Allocations Portal
 - <https://allocations.access-ci.org>
- ACCESS User Support:
 - <https://support.access-ci.org>
- ACCESS Documentation:
 - <https://docs.access-ci.org>
- SDSC HPC User Support
 - consult@sdsc.edu
- SDSC Documentation
 - <https://sdsc.edu/>



Questions

- SDSC Support
 - consult@sdsc.edu
- ACCESS Allocations Portal: ACCESS Allocations Portal
 - <https://allocations.access-ci.org>
- ACCESS User Support: ACCESS Support
 - <https://support.access-ci.org>
- ACCESS Documentation: ACCESS Documentation
 - <https://docs.access-ci.org>

